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Multilingual cyberbullying detection

Advances, Challenges, and Future Directions in Multilingual Cyberbullying Detection

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Transformer-based and Hybrid Models	Transformer architectures (e.g., mBERT, XLM-RoBERTa, mT5) and hybrid models (e.g., transformer-BiLSTM) significantly improve multilingual cyberbullying detection, especially when augmented with emotion-aware and severity-aware features.	1 2 3 4 5
Hybrid Ensemble & Multimodal Approaches	Hybrid systems combining classical ML, deep learning, and transformer embeddings, as well as multimodal frameworks (text, audio, image, video), outperform unimodal models and capture cultural cues.	6 7 8
Dataset & Annotation Challenges	Datasets are often imbalanced, inconsistently annotated, and underrepresent low-resource/code-mixed languages. Synthetic data and LLM-based augmentation are promising but require careful evaluation.	9 10 11 12
Cultural & Linguistic Nuances	Embedding dialectal, cultural, and linguistic features (e.g., dialect-specific pretraining, emotion/sentiment analysis) improves accuracy and generalizability. Explainable AI (SHAP, LIME) is critical for transparency.	13 14 15 16
Privacy & Scalability	Federated learning frameworks, combined with explainable AI, address privacy and scalability, enabling collaborative, privacy-preserving, and interpretable detection.	17 18 19 20

Direct Answer

Multilingual cyberbullying detection research has evolved by leveraging advanced transformer-based models and hybrid ensemble and multimodal approaches to tackle the linguistic, cultural, and data scarcity challenges inherent in detecting abusive content across languages. Researchers have integrated emotion-aware and severity-aware frameworks to better capture nuanced cyberbullying behaviors, while new federated learning and explainable AI methodologies address critical privacy and interpretability concerns. However, significant research gaps remain in standardizing annotations, balancing datasets, and ensuring scalability and transparency in low-resource and code-mixed language settings.

Study Scope

Time Period: Primarily 2018–2024

Disciplines: Natural Language Processing, Machine Learning, Computational Social Science, Ethics in AI

Methods: Transformer-based models, hybrid/ensemble learning, multimodal fusion, semi-supervised learning, federated learning, explainable AI, data augmentation, annotation studies

Assumptions & Limitations

Most findings are based on social media datasets, which may not generalize to all online platforms.

Low-resource and code-mixed languages remain underrepresented in both datasets and model evaluations.

Annotation standards and cultural context vary widely, affecting cross-study comparability.

Synthetic data and LLM-based augmentation are promising but require further validation for real-world deployment.

Suggested Further Research

Standardize annotation protocols across languages and dialects.

Develop scalable, interpretable hybrid models integrating federated learning and explainable AI.

Expand multimodal datasets and models, especially for underrepresented languages and modalities.

Address ethical considerations, including privacy, fairness, and dialectal bias mitigation.

1. Introduction

Background and Motivation

The proliferation of social media and digital communication platforms has led to a surge in cyberbullying incidents worldwide, transcending linguistic and cultural boundaries. The complexity of online abuse is amplified in multilingual and code-mixed environments, where linguistic diversity, informal language, and cultural nuances challenge traditional detection systems. As online abuse increasingly targets users across languages and regions, there is a critical need for robust, culturally aware, and scalable cyberbullying detection systems that can operate effectively in multilingual contexts. Recent advances in natural language processing (NLP), particularly transformer-based models and hybrid multimodal frameworks, have shown promise in addressing these challenges, but significant gaps remain in data quality, annotation consistency, interpretability, and privacy preservation [1] [21] [22].

2. Methodologies for Multilingual Cyberbullying Detection

2.1 Transformer-Based and Cross-Lingual Models

Transformer architectures such as BERT, XLM-RoBERTa, and mT5 have become the backbone of state-of-the-art multilingual cyberbullying detection systems. These models excel at capturing contextual and semantic nuances across languages, outperforming traditional machine learning and earlier deep learning models [1] [2] [3].

Sequence-to-sequence models (e.g., mT5) and hybrid transformer-BiLSTM architectures offer a balance between accuracy and computational efficiency, with hybrid models reducing training time by over 55% and increasing inference speed by more than 80% compared to pure transformers, while maintaining or improving accuracy [4] [23].

Dialect-specific transformers (e.g., MARBERT, AraBERT) outperform general multilingual models in dialect-rich languages such as Arabic, highlighting the importance of in-domain pretraining and context-aware annotation [24].

Synthesis: Transformer-based models, especially when adapted or hybridized for specific dialects or tasks, are the current gold standard for multilingual cyberbullying detection, but computational cost and interpretability remain challenges [25].

2.2 Hybrid and Ensemble Approaches

Hybrid models combine transformer embeddings with classical classifiers (e.g., SVM, Logistic Regression) or sequential models (e.g., BiLSTM), achieving high accuracy and improved interpretability, especially in code-mixed and low-resource settings [6] [8].

Ensemble strategies leverage the strengths of multiple models, including CNNs, RNNs, and transformers, to enhance robustness and generalization across diverse languages and modalities [26] [27].

Explainability is enhanced in hybrid models through integration with SHAP, LIME, and attention visualization, supporting transparency and trust in multilingual contexts [25] [28].

Synthesis: Hybrid and ensemble approaches are particularly effective for code-mixed and low-resource languages, balancing accuracy, interpretability, and computational efficiency [8].

2.3 Semi-Supervised and Low-Resource Strategies

Semi-supervised learning frameworks, such as confidence-guided pseudo-labeling and self-training, leverage both labeled and unlabeled data to boost performance in low-resource languages, achieving up to 94% accuracy in Bangla and Chittagonian [22] [29].

Data augmentation (e.g., synonym replacement, back-translation, paraphrasing) and transfer learning from high-resource to low-resource languages improve minority class detection and overall robustness [11] [30].

Dialect-specific pretraining (e.g., DarijaBERT for Moroccan Arabic) is crucial for capturing subtle forms of cyberbullying and improving detection in under-resourced settings [30].

Synthesis: Semi-supervised and transfer learning strategies are essential for addressing data scarcity and linguistic diversity in multilingual cyberbullying detection [31].

2.4 Multimodal Detection Frameworks

Multimodal frameworks integrate text, audio, image, and video modalities using advanced fusion strategies (early, late, hierarchical) and pretrained models (e.g., CLIP, XLM-RoBERTa, ViT, GNNs) to capture subtle cues such as sarcasm, irony, and implicit abuse [7] [32] [33].

Multitask learning incorporating sentiment, emotion, and sarcasm detection as auxiliary tasks further enhances performance, especially in meme and code-mixed datasets [32] [34].

Operational stability is maintained through techniques like zero-vector substitution for missing modalities, enabling real-time, scalable detection [32].

Synthesis: Multimodal frameworks are increasingly necessary for real-world cyberbullying detection, as online abuse often spans text, images, and audio, especially in multilingual and code-mixed environments [35].

2.5 Explainability and Interpretability

Explainable AI (XAI) methods such as SHAP, LIME, and attention mechanisms are integrated into detection systems to provide rationale identification, sentiment analysis, and transparent decision-making [25] [28] [36].

Multitask models that jointly predict cyberbullying, sentiment, target groups, and rationales improve both performance and interpretability, particularly in code-mixed and low-resource languages [37].

Synthesis: Explainability is a growing focus, driven by regulatory requirements and the need for human-in-the-loop moderation, especially in linguistically and culturally complex settings [25].

3. Datasets and Annotation Standards

3.1 Multilingual and Code-Mixed Datasets

Large and diverse datasets include Malay-English (52,140 instances), Bangla-Chittagonian (10,000+), Tamil (47,692), and multi-domain datasets covering Hindi, Marathi, Arabic, French, German, Spanish, and more [9] [29] [38] [39].

Code-mixed datasets are annotated for multiple labels (e.g., cyberbullying, sentiment, emotion, target group) and often include severity levels and user profiling [9] [40].

Multimodal datasets (e.g., memes, audio, video) reflect the complexity of real-world cyberbullying and support advanced detection frameworks [7] [41].

Synthesis: Dataset diversity is increasing, but low-resource and code-mixed languages remain underrepresented, and multimodal data is still emerging [9].

3.2 Annotation Standards and Practices

Annotation schemes for code-mixed datasets are multi-layered, capturing language switching, pragmatic roles, and severity, while monolingual datasets use more standardized protocols [42] [43].

Quality control involves inter-annotator agreement metrics (e.g., Cohen's Kappa, Fleiss' Kappa, Krippendorff's alpha) and expert supervision [44] [45].

Privacy-aware annotation includes anonymization of personal information and session-based labeling for abuser/victim roles [43].

Synthesis: Annotation standards are evolving toward greater granularity and privacy awareness, but inconsistency and subjectivity remain challenges, especially in code-mixed and dialect-rich datasets [46].

3.3 Preprocessing and Normalization Techniques

Effective preprocessing includes URL, emoji, and stop-word removal, tokenization, lemmatization, stemming, and normalization for transliteration and spelling variation [28] [47].

Lexical normalization using byte-level language models (e.g., ByT5) and expansion of abbreviations improves robustness in noisy social media text [48] [49].

Strict preprocessing is essential for code-mixed data to enable effective use of multilingual transformers [28].

Synthesis: Preprocessing and normalization are foundational for high-accuracy detection, especially in informal, code-mixed, and low-resource language settings [50].

3.4 Synthetic Data Generation and LLM-Based Augmentation

LLM-based augmentation and synthetic data generation are cost-effective for addressing data scarcity, enabling classifiers to approach the performance of models trained on authentic data [12] [51].

Data augmentation (e.g., synonym replacement, back-translation) improves minority class detection, though gains in overall accuracy may be modest [11].

Prompt engineering and auxiliary tasks (e.g., aggression detection) further enhance LLM-based detection [52].

Synthesis: Synthetic data and LLM augmentation are promising for balancing datasets and improving low-resource language coverage, but require careful evaluation to ensure diversity and realism [12].

4. Cultural and Linguistic Nuances in Detection

4.1 Impact of Cultural and Linguistic Diversity

Cultural and linguistic differences significantly affect detection accuracy, requiring models to integrate rule-based features, contextual neural modeling, and metaheuristic optimization [21] [53].

Low-resource and code-mixed languages present challenges due to scarce annotations, informal language, and dialectal variability [30] [54].

Emotion-aware and severity-aware frameworks improve detection by capturing emotional context and cultural expressions [38] [55].

Synthesis: Addressing cultural and linguistic diversity is essential for accurate and generalizable cyberbullying detection, especially in multilingual and code-mixed environments [53].

4.2 Integration of Cultural and Linguistic Features

Emotion-aware features (e.g., from DistilBERT, BiLSTM) and sentiment analysis significantly boost detection accuracy, especially in low-resource languages [5] [22].

Dialect-specific pretraining (e.g., AraBERT, MARBERT) and rule-based feature extraction (e.g., negation, intensifiers) enhance interpretability and performance [21] [56].

Multi-input integrative learning processes language-specific and typographic cues separately before fusion, improving real-time detection in code-mixed data [13].

Synthesis: Incorporating emotion, sentiment, and dialectal features is critical for culturally aware and effective multilingual cyberbullying detection [57].

4.3 Emotion-Aware and Severity-Aware Frameworks

Emotion-aware frameworks achieve F1-scores up to 0.97 by capturing subtle emotional triggers, while severity-aware models enable targeted interventions and reach up to 98.42% accuracy [5] [9].

Multitask learning with sentiment, emotion, and sarcasm detection improves performance by over 3% in accuracy and F1-score [58].

Synthesis: Emotion and severity awareness are key to nuanced, context-sensitive detection, especially in multilingual and multimodal settings [27].

4.4 Explainability in Code-Mixed and Multilingual Detection

Explainability techniques (SHAP, LIME, attention mechanisms) provide both global and local explanations, improving transparency and trust [14] [36].

Human-annotated rationales and attention visualization enhance interpretability, especially in code-mixed and low-resource languages [59].

Synthesis: Explainability is essential for regulatory compliance, human moderation, and ethical deployment in linguistically complex environments [60].

4.5 Dialect-Specific Versus General Multilingual Models

Dialect-specific models (e.g., MARBERT, AraBERT) consistently outperform general multilingual transformers in dialect-rich languages, achieving higher F1-scores and accuracy [24].

Hybrid transformer models combining dialect-specific and general models further enhance performance [61].

Synthesis: Dialect-specific pretraining and hybridization are crucial for high-accuracy detection in dialectally diverse languages [62].

5. Research Gaps and Future Directions

5.1 Current Gaps and Challenges

Data scarcity in low-resource and code-mixed languages, **annotation inconsistency**, and **class imbalance** remain major obstacles [22] [53] [56].

Lack of interpretability and **privacy concerns** limit real-world deployment and user trust [25] [63].

Dataset imbalance is primarily due to the prevalence of majority classes and limited annotated data for minority classes [64] [65].

Synthesis: Addressing these gaps requires methodological innovation, standardization, and ethical awareness [22].

5.2 Data Imbalance and Annotation Inconsistency

Oversampling (SMOTE, Random Oversampling) and **data augmentation** (synonym replacement, back-translation) are effective for minority class improvement, but no single technique is universally superior [11] [66].

Annotation inconsistency due to dialectal diversity and informal language reduces dataset reliability and model performance; involving native speakers and majority voting improves quality [22] [67].

Synthesis: Systematic preprocessing, normalization, and community-driven annotation are essential for reliable multilingual datasets [28].

5.3 Federated Learning and Explainable AI for Privacy and Transparency

Federated learning (FL) enables collaborative model training without sharing raw data, preserving privacy [17] [68].

Explainable AI (SHAP, LIME) can be integrated with FL to provide transparent, interpretable models while maintaining data decentralization [15] [19].

Blockchain integration further enhances security and auditability in federated frameworks [69].

Synthesis: Combining FL and XAI is a promising direction for privacy-preserving, transparent, and scalable multilingual cyberbullying detection [18].

5.4 Hybrid Supervised-Unsupervised and Multimodal Models

Hybrid models combining supervised, unsupervised, and multimodal learning improve scalability, robustness, and contextual understanding [70] [71].

Multi-task self-training and **feature fusion** enhance performance in multi-label and multilingual scenarios [71] [72].

Synthesis: Future systems should integrate reasoning over user intent, pragmatics, and conversational context for adaptive, real-time moderation [72].

5.5 Ethical Frameworks and Bias Mitigation

Ethical annotation involves diverse, culturally aware annotators and community engagement to reduce bias [73] [74].

Dialectal bias mitigation uses balanced multilingual training, counterfactual data augmentation, and fairness-aware learning [75] [76].

Transparency is supported by explainable AI and controlled evaluation frameworks [77] [78].

Synthesis: Ethical, fairness-aware, and community-driven frameworks are essential for equitable and trustworthy multilingual cyberbullying detection [79].

6. Real-World Implications

Deployment: Multilingual and multimodal detection systems are increasingly integrated into social media moderation tools, but require ongoing adaptation to new languages, dialects, and abuse modalities [7].

Policy: Explainable and privacy-preserving models support regulatory compliance (e.g., GDPR) and user trust [15] [18].

Ethics: Community-driven annotation and fairness-aware modeling help mitigate bias and ensure culturally sensitive moderation [73] [79].

Scalability: Federated learning and hybrid models enable scalable, collaborative detection across platforms and regions [17] [71].

7. Future Research Directions

Standardize annotation protocols across languages, dialects, and modalities to improve dataset reliability and cross-study comparability [74] [80].

Develop scalable, interpretable hybrid models that integrate federated learning, explainable AI, and multimodal fusion for robust, privacy-preserving detection [18] [71].

Expand multimodal datasets and models for underrepresented languages and content types (e.g., memes, audio, video) [7] [41].

Advance ethical frameworks for annotation and model development, focusing on dialectal bias mitigation, transparency, and community engagement [73] [79].

Explore unified, dynamic platforms that aggregate dialectal variations in real time, interface with moderation tools, and leverage blockchain for secure, transparent data sharing [creative_insights].

Synthesis

Recent advancements in multilingual cyberbullying detection demonstrate the effectiveness of transformer-based and hybrid ensemble models that integrate multimodal data, sentiment analysis, and culturally aware features to capture complex abusive behaviors across languages. However, challenges persist in dataset imbalance, annotation consistency, and model interpretability, particularly for low-resource and code-mixed languages. Emerging approaches such as federated learning coupled with explainability techniques offer promising avenues for privacy-preserving, scalable solutions, yet further research is essential to develop standardized, ethically sound, and highly interpretable detection systems [1] [7] [9] [13] [17].

References

1. Comparative Analysis of Cyberbullying Detection: A case study for Turkish and English Najib, A., Ozel, S.A., Coban, O. 2023 Innovations in Intelligent Systems and Applications Conference, ASYU 2023, 2023 https://www.scopus.com/pages/publications/85178268519?origin=scopusAI

2. A Text-to-Text Model for Multilingual Offensive Language Identification Ranasinghe, T., Zampieri, M. IJCNLP-AACL 2023 - 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, Findings of the Association for Computational Linguistics: IJCNLP-AACL 2023, 2023 https://www.scopus.com/pages/publications/85188550806?origin=scopusAI

3. Unified Transformer Framework for Automated Cyberbullying Detection Alikhashashneh, E., Almomani, A., Alsawan, H., (...), Pan, V.S.-H. International Journal of Cloud Applications and Computing, 2025 https://www.scopus.com/pages/publications/105012382813?origin=scopusAI

4. Enhancing Chatbot Responses through Improved T5 Model Incorporating Aggregated Multi-Head Attention Mechanism and Bidirectional Long Short-Term Memory Muthukumaran, N., Vignesh, A. Journal of Universal Computer Science, 2025 https://www.scopus.com/pages/publications/105012546079?origin=scopusAI

5. Emotion-Driven Cyberbullying Detection: A Hybrid Deep Learning Approach Integrating CNN-LSTM and BERT for Enhanced Accuracy Lakkshmanan, A., Kandadi, R.R., Manda, S.A. Proceedings of the 9th International Conference on Communication and Electronics Systems, ICCES 2024, 2024 https://www.scopus.com/pages/publications/85218408669?origin=scopusAI

6. Multi-Model Ensemble Approach for Enhanced Cyberbullying Detection Across Diverse Categories Patil, S., Punde, S., Sahani, P., Salve, A. Proceedings of 5th International Conference on IoT Based Control Networks and Intelligent Systems, ICICNIS 2024, 2024 https://www.scopus.com/pages/publications/85217170256?origin=scopusAI

7. Multilingual multimodal cyberbullying detection through adaptive and hierarchical fusion Ismail, W.S., Ullah, H., Adnan, M., Ullah, F. Array, 2026 https://www.scopus.com/pages/publications/105028782013?origin=scopusAI

8. An Ensemble Framework Leveraging BERT and SVM for Effective Multiclass Cyberbullying Detection Sharma, P., Dubey, A.K., Arjunkumar, G., (...), Nandan, D. Lecture Notes in Electrical Engineering, 2026 https://www.scopus.com/pages/publications/105031491382?origin=scopusAI

9. CYBERBULLYING SEVERITY DETECTION IN MALAY-ENGLISH CODE-MIXED TEXT: A HYBRID NLP-CNN APPROACH WITH USER BEHAVIOR PROFILING Inusa, I., Anwar, R.M., Abu Bakar, A., (...),

Latif, A.A. Journal of Theoretical and Applied Information Technology, 2026 https://www.scopus.com/pages/publications/105042496816?origin=scopusAI

10. Arabic Cyberbullying Detection: A Comprehensive Review of Datasets and Methodologies Aljalaoud, H., Dashtipour, K., Al-Dubai, A.Y. IEEE Access, 2025 https://www.scopus.com/pages/publications/105003756707?origin=scopusAI

11. Enhancing Bangla Cyberbullying Detection with Data Augmentation: A Performance Evaluation Ul Anam, M.M., Abrar, A., Haque, M.M., (...), Amin, M.A. 2025 IEEE 23rd Student Conference on Research and Development, SCOReD 2025 - Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105035746298?origin=scopusAI

12. A Multi-Aspect Evaluation Framework for Synthetic Data: Case Study on Irony and Sarcasm Majer, L., Barić, A., Sandalj, F., (...), Šnajder, J. WASSA 2026 - 15th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis, Proceedings of the Workshop, 2026 https://www.scopus.com/pages/publications/105040196499?origin=scopusAI

13. Multi-input integrative learning using deep neural networks and transfer learning for cyberbullying detection in real-time code-mix data Kumar, A., Sachdeva, N. Multimedia Systems, 2022 https://www.scopus.com/pages/publications/85087757952?origin=scopusAI

14. Evaluating explainability in language classification models: A unified framework incorporating feature attribution methods and key factors affecting faithfulness Dehdarirad, T. Data and Information Management, 2025 https://www.scopus.com/pages/publications/105000645997?origin=scopusAI

15. Analyzing the Role of LIME and SHAP in Explainable DoS Attack Detection for IoT Systems Paul, A., Kumari, S., Navadia, N.R., Sinha, S. 4th International Conference on Automation, Computing and Renewable Systems, ICACRS 2025 - Proceedings, 2025 https://www.scopus.com/pages/publications/105032756418?origin=scopusAI

16. Explainability and Interpretability of Multilingual Large Language Models: A Survey Resck, L., Augenstein, I., Korhonen, A. EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Conference, 2025 https://www.scopus.com/pages/publications/105040271188?origin=scopusAI

17. FedCMB: A Cross-Modal Federated Learning Framework for Multi-Modal Cyberbullying Detection with Shared Feature Space Zhang, W., Dong, C., Wang, P. Proceedings - 2025 IEEE 10th International Conference on Data Science in Cyberspace, DSC 2025, 2025 https://www.scopus.com/pages/publications/105032998969?origin=scopusAI

18. Federated SHAP: Privacy-Preserving and Consistent Post-hoc Explainability in Federated Learning Ducange, P., Marcelloni, F., Miglionico, G.C., (...), Ruffini, F. Machine Learning, 2026 https://www.scopus.com/pages/publications/105027956425?origin=scopusAI

19. Implementing Federated Learning and Explainability Techniques in Regression Models to Increase Transparency and Reliability Ribeiro, J., Santos, R., Analide, C., Silva, F. Studies in Informatics and Control, 2024 https://www.scopus.com/pages/publications/105013482710?origin=scopusAI

20. Federated Learning and Privacy-Preserving Techniques: A Comprehensive Review of Advancements and Challenges Reshma, Keyal, P.K., Saini, S.S. International Conference on Emerging Technologies in Electronics and Green Energy, ICETEG 2025, 2025 https://www.scopus.com/pages/publications/105037574520?origin=scopusAI

21. An Integrated Linguistic and Metaheuristic-Optimized Elman Neural Network Framework for Cyberbullying Detection Aisyah, S., Sembiring, A., Faadhil, (...), Zuhanda, M.K. Journal of Applied Data Sciences, 2026 https://www.scopus.com/pages/publications/105044892562?origin=scopusAI

22. Emotion-Aware Semi-Supervised Framework for Cyberbullying and Emotion Detection in Low-Resource Bangla and Chittagonian Texts Al Noman, A., Sultana, T., Arzoo, S.A., (...), Sabbir, A. 2026 International

Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure, PECCII 2026, 2026 https://www.scopus.com/pages/publications/105049676143?origin=scopusAI

23. A Resource-efficient Text-to-Text Transfer Transformer Encoder-based Vertical Hybrid Model for Malicious URLs Detection Zhao, Z., Chen, J., Messou, F.J.A., (...), Yu, K. IEEE Vehicular Technology Conference, 2024 https://www.scopus.com/pages/publications/85213071857?origin=scopusAI

24. TRANSFORMER-BASED CYBERBULLYING DETECTION IN MOROCCAN DARIJA: A WILLARD TAXONOMY APPROACH Tahiri, A., Sbihi, B., Gouarti, R., (...), El Ksimi, A. Journal of Theoretical and Applied Information Technology, 2026 https://www.scopus.com/pages/publications/105039513057?origin=scopusAI

25. Explainable Multilingual Hybrid Transformer-Based Model for Cyberbullying Detection in Social Media: A Systematic Review Azeem, K., Arshad, F., Wasim, M. 2026 International Conference on Information Science and Communication Technology, ICISCT 2026, 2026 https://www.scopus.com/pages/publications/105044301428?origin=scopusAI

26. Detection of Cyberbully in Social Media Platform Using Deep Learning Algorithm Ruby, S., Sudha, S. 2026 International Conference on Innovations in Emerging Technologies for Sustainable Development: Empowering Innovation for a Sustainable Future, ICIETSD 2026 - Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105045454312?origin=scopusAI

27. Hybrid 1D VGG16 and SVM Framework for Early Detection and Intensity Classification of Cyberbullying Sahana, V., Kumar, A.K.M. International Journal of Intelligent Engineering and Systems, 2025 https://www.scopus.com/pages/publications/105000992836?origin=scopusAI

28. ExBERT-CM: Explainable Cyberbullying Detection in Code-Mixed Social Media with Multilingual Transformer Models Varshini, R., Priyanga Mary, J., Shoba, R. 2025 IEEE 9th International Conference on Information and Communication Technology, CICT 2025, 2025 https://www.scopus.com/pages/publications/105041763287?origin=scopusAI

29. BanCyB: a hybrid transformer-LSTM framework for explainable multi-label cyberbullying detection in low-resource setting Hasan, M.M., Arefin, M.S., Mia, M.A., Dhar, P.K. Discover Applied Sciences, 2026 https://www.scopus.com/pages/publications/105038893614?origin=scopusAI

30. Leveraging Transformer-Based Models for Cyberbullying Detection in the Moroccan Arabic Dialect Laachari, H., Kobbane, A., Tembine, H. International Conference on Wireless Networks and Mobile Communications, WINCOM, 2025 https://www.scopus.com/pages/publications/105034194941?origin=scopusAI

31. Semi-supervised learning for cyberbullying detection in social networks Nahar, V., Al-Maskari, S., Li, X., Pang, C. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2014 https://www.scopus.com/pages/publications/84904205580?origin=scopusAI

32. Advanced Multimodal Multitask Framework for Cyberbullying Detection Deshmukh, D., Parasar, D., Kukreja, S., Shukla, G. IET Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105040196928?origin=scopusAI

33. Multi-Model Fusion for Enhanced Cyber-Bullying Detection in Digital Communication Sree, S.S., Joseph, N.L. CINS 2024 - 2nd International Conference on Computational Intelligence and Network Systems, 2024 https://www.scopus.com/pages/publications/85219520404?origin=scopusAI

34. Multimodal Cyberbullying Detection Using Inter-Modal-Attention-Based Deep Learning Suneetha, M., Rajesh, T., Gopinath, P.L., (...), Reddy, K.V.N. IEEE International Conference on Innovate for Humanitarian: Tech Solutions for Global Challenge, ICIH 2025, 2025 https://www.scopus.com/pages/publications/105046425129?origin=scopusAI

35. Cyberbullying Detection in Social Media with Multimodal Data Using Transfer Learning Valliyammai, C., Raaj, S.M., Athish, B.L., Kumar, J.K. Proceedings of 2024 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering, WIECON-ECE 2024, 2024 https://www.scopus.com/pages/publications/105001394285?origin=scopusAI

36. “Explain Thyself Bully”: Sentiment Aided Cyberbullying Detection with Explanation Maity, K., Jha, P., Jain, R., (...), Bhattacharyya, P. Lecture Notes in Computer Science, 2023 https://www.scopus.com/pages/publications/85173581672?origin=scopusAI

37. Towards Safer Digital Spaces: Automated Detection of Cyberbullying Through Multi-Modal Learning Karthikeyan, M.S., Francis Raj, J.R., Krishnan, R.S., (...), Joshua, K.P. 5th International Conference on Sustainable Communication Networks and Application, ICSCNA 2024 - Proceedings, 2024 https://www.scopus.com/pages/publications/85219612370?origin=scopusAI

38. Multi-tier linguistic and emotional modeling for cyberbullying detection in Tamil social media V., J.P., S., A.A.V. Expert Systems with Applications, 2026 https://www.scopus.com/pages/publications/105013738362?origin=scopusAI

39. A multilingual, multimodal dataset of aggression and bias: the ComMA dataset Kumar, R., Ratan, S., Singh, S., (...), Bansal, A. Language Resources and Evaluation, 2024 https://www.scopus.com/pages/publications/85176761347?origin=scopusAI

40. A Multitask Multimodal Framework for Sentiment and Emotion-Aided Cyberbullying Detection Maity, K., Kumar, A., Saha, S. IEEE Internet Computing, 2022 https://www.scopus.com/pages/publications/85126274162?origin=scopusAI

41. Multimodal and Multilingual Cyberbullying Detection Using Deep Learning Jagannadhachari, D., Kumar, A., Deepika, S., (...), Agrawal, A. Lecture Notes in Networks and Systems, 2025 https://www.scopus.com/pages/publications/105006822976?origin=scopusAI

42. Creating a large Multi-Layered representational repository of Linguistic Code Switched Arabic data Diab, M., Ghoneim, M., Hawwari, A., (...), Al-Badrashiny, M. Proceedings of the 10th International Conference on Language Resources and Evaluation, LREC 2016, 2016 https://www.scopus.com/pages/publications/85037143873?origin=scopusAI

43. Privacy-Aware Code-Mixed Cyberbullying Dataset for Session-Based Analysis Chu, C.C.F., Tong, C.C.H., Chiu, C.H., (...), Lam, S.C. Data, 2026 https://www.scopus.com/pages/publications/105034631591?origin=scopusAI

44. SD-CVD Corpus: Towards Robust Detection of Fine-Grained Cyber-Violence Across Saudi Dialects in Online Platforms Alsayed, A., Elhag, S., Badri, S. Information (Switzerland), 2026 https://www.scopus.com/pages/publications/105028560614?origin=scopusAI

45. Expert-Annotated Dataset to Study Cyberbullying in Polish Language Ptaszynski, M., Pieciukiewicz, A., Dybala, P., (...), Wroczynski, M. Data, 2024 https://www.scopus.com/pages/publications/85183934774?origin=scopusAI

46. Two languages, one treebank: building a Turkish–German code-switching treebank and its challenges Çetinoğlu, Ö., Çöltekin, Ç. Language Resources and Evaluation, 2023 https://www.scopus.com/pages/publications/85124847200?origin=scopusAI

47. AI-Driven Cyberbullying Detection in Online Communication: Leveraging Machine Learning and Natural Language Processing for Safer Digital Interactions Kalidas, P., Malini, S., Latika, S., Sri, D.K. 2025 IEEE 1st International Conference on Innovations in Engineering and Next-Generation Technologies for Sustainability, ICINVENTS 2025, 2025 https://www.scopus.com/pages/publications/105036966784?origin=scopusAI

48. ÚFAL at MultiLexNorm 2021: Improving Multilingual Lexical Normalization by Fine-tuning ByT5 Samuel, D., Straka, M. W-NUT 2021 - 7th Workshop on Noisy User-Generated Text, Proceedings of the Conference, 2021 https://www.scopus.com/pages/publications/85117780247?origin=scopusAI

49. Improving English Text Normalization With Abbreviation Expansion Kawata, N., Kurokawa, Y., Fukushi, M. 2026 IEEE 2nd International Conference on Consumer Technology, ICCT-Pacific 2026, 2026 https://www.scopus.com/pages/publications/105041904239?origin=scopusAI

50. An Efficient Hybrid Deep Learning Framework for Cyberbullying Detection Durgekar, S.R., Hegde, R., Hegde, S.K. Proceedings of 2nd International Conference on Multi-Agent Systems for Collaborative Intelligence, ICMSCI 2026, 2026 https://www.scopus.com/pages/publications/105037941789?origin=scopusAI

51. Synthetic vs. Gold: The Role of LLM Generated Labels and Data in Cyberbullying Detection Kazemi, A., Kalaivendan, S.B.N., Wagner, J., (...), Davis, B. International Conference Recent Advances in Natural Language Processing, RANLP, 2025 https://www.scopus.com/pages/publications/105034180687?origin=scopusAI

52. Cyberbullying Detection via Aggression-Enhanced Prompting Saeid, A., Sabu, A., Koushik, G.A., (...), Kanojia, D. International Conference Recent Advances in Natural Language Processing, RANLP, 2025 https://www.scopus.com/pages/publications/105034085614?origin=scopusAI

53. Linguistic Challenges in Cyberbullying Detection for Code-Mixed Languages: A Review Jadav, N., Patel, M., Jajal, B. Procedia Computer Science, 2026 https://www.scopus.com/pages/publications/105045579841?origin=scopusAI

54. Cyberbullying Detection in Low Resource Code-Mixed Languages Using ML and NLP Techniques Saralaya, N., Jashma Suresh, P.P., Saralaya, S. Lecture Notes in Networks and Systems, 2025 https://www.scopus.com/pages/publications/105010183625?origin=scopusAI

55. Multilingual and Severity-Aware Cyberbullying Detection with Emotion Fusion and Transformer Models Vetal, N., Vasgi, B. 2025 IEEE Pune Section International Conference, PuneCon 2025, 2025 https://www.scopus.com/pages/publications/105034656866?origin=scopusAI

56. Cyberbullying detection approaches for Arabic texts: a systematic literature review Allwaibed, H., Anbar, M., Manickam, S., Bintang, A. Frontiers in Artificial Intelligence, 2025 https://www.scopus.com/pages/publications/105020589892?origin=scopusAI

57. Global Texts, Unified Action: Tackling Cyberbullying Across Multiple Languages Faraj, A., Utku, S. International Journal of Pattern Recognition and Artificial Intelligence, 2024 https://www.scopus.com/pages/publications/85204126047?origin=scopusAI

58. A Multitask Framework for Sentiment, Emotion and Sarcasm aware Cyberbullying Detection from Multi-modal Code-Mixed Memes Maity, K., Jha, P., Saha, S., Bhattacharyya, P. SIGIR 2022 - Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022 https://www.scopus.com/pages/publications/85135033392?origin=scopusAI

59. X-MuTeST: A Multilingual Benchmark for Explainable Hate Speech Detection and A Novel LLM-consulted Explanation Framework Rehman, M.Z.U., Kasu, S.K.R., Koppula, S.R., (...), Kumar, N. Proceedings of the AAAI Conference on Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105035094956?origin=scopusAI

60. Simplifying Outcomes of Language Model Component Analyses with ELIA Eidt, A.L., Feldhus, N. EACL 2026 - 19th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference, Vol. 1 - (Long Papers), 2026 https://www.scopus.com/pages/publications/105040068407?origin=scopusAI

61. Leveraging Transformers for Detection of Arabic Cyberbullying on Social Media: Hybrid Arabic Transformers Alsuwaylimi, A.A., Alenezi, Z.S. Computers, Materials and Continua, 2025 https://www.scopus.com/pages/publications/105003261430?origin=scopusAI

62. ARBERT & MARBERT: Deep bidirectional transformers for Arabic Abdul-Mageed, M., Elmadany, A., Nagoudi, E.M.B. ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference, 2021 https://www.scopus.com/pages/publications/85116612882?origin=scopusAI

63. Cyberbullying Detection on Social Networks, Opportunities and Future Challenges: A Systematic Literature Review Tipantiza, N., Quishpe, A., Fuertes, W., Andrade, R. Smart Innovation, Systems and Technologies, 2026 https://www.scopus.com/pages/publications/105027304222?origin=scopusAI

64. LLM-based text plus emoji multiclass hate speech language detection for resource constrained devices Anjum, N., Latif, Z., Chen, H. Knowledge-Based Systems, 2026 https://www.scopus.com/pages/publications/105034725961?origin=scopusAI

65. Performance Evaluation of Text Augmentation Methods with BERT on Small-sized, Imbalanced Datasets Hu, L., Li, C., Wang, W., (...), Shang, Y. Proceedings - 2022 IEEE 4th International Conference on Cognitive Machine Intelligence, CogMI 2022, 2022 https://www.scopus.com/pages/publications/85150679486?origin=scopusAI

66. SMOTE-NGO of Stacking Ensemble Cyberbullying Detection of Feature and Model Parameter Optimization Abdalrdha, Z.K., Hashim, M.S., Mutar, J.R., Dawood, S.H. Ingenierie des Systemes d'Information, 2026 https://www.scopus.com/pages/publications/105045256815?origin=scopusAI

67. KuHateCorpus: Annotated Kuwaiti Dialect Dataset for Hate Speech Detection in Cybersecurity Education Al-Shammari, E.T. ACM SIGCITE 2025 - Proceedings of the 26th ACM Annual Conference on Cybersecurity and Information Technology Education, 2025 https://www.scopus.com/pages/publications/105028734721?origin=scopusAI

68. Towards federated learning: An overview of methods and applications Silva, P.R., Vinagre, J., Gama, J. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 2023 https://www.scopus.com/pages/publications/85146177076?origin=scopusAI

69. FedBGS: A Blockchain Approach to Segment Gossip Learning in Decentralized Systems Turazza, F., Pietri, M., Picone, M., Mamei, M. Proceedings - 2025 IEEE 45th International Conference on Distributed Computing Systems Workshops, ICDCSW 2025, 2025 https://www.scopus.com/pages/publications/105030445516?origin=scopusAI

70. A Hybrid BERT-CNN with Multihead Self-Attention for Automated Cyberbullying Detection Yerekesheva, M., Akhmetova, O., Kaziyeva, A., (...), Abdimukhan, T. Engineering, Technology and Applied Science Research, 2026 https://www.scopus.com/pages/publications/105037716166?origin=scopusAI

71. HMS-BERT: Hybrid Multi-Task Self-Training for Multilingual and Multi-Label Cyberbullying Detection Feng, Z., Cui, X., Sun, Y., (...), Hasan, M.M. 2026 9th International Conference on Artificial Intelligence and Big Data, ICAIBD 2026, 2026 https://www.scopus.com/pages/publications/105048323256?origin=scopusAI

72. Breaking the classification bubble: Rethinking cyberbullying detection and detoxification as knowledge-guided reasoning systems Nadeem, M., Hongsong, C. Computer Science Review, 2026 https://www.scopus.com/pages/publications/105041027133?origin=scopusAI

73. Socio-Economic Diversity in Human Annotations Fan, S., Barlas, P., Christoforou, E., (...), Demartini, G. ACM International Conference Proceeding Series, 2022 https://www.scopus.com/pages/publications/85133677478?origin=scopusAI

74. Hybrid Ontology and NLP-Based Annotation Model for Indigenous Lexicon and Folklore Preservation Khudayberganova, D., Tosheva, N., Aslanova, H., (...), Tursunova, D. Indian Journal of Information Sources and Services, 2025 https://www.scopus.com/pages/publications/105010078614?origin=scopusAI

75. Mitigating Cultural Bias for Low-Resource Languages in Multilingual Large Language Models Binegde, G.N., Zhang, H., Li, Q., (...), Yan, R. Data Intelligence, 2026 https://www.scopus.com/pages/publications/105044240720?origin=scopusAI

76. Fairness-Aware Few-Shot Learning for Audio-Visual Stress Detection Shelke, A.S., Sneh, A., Adyasha, A., Lone, H.R. IEEE Transactions on Affective Computing, 2026 https://www.scopus.com/pages/publications/105050236798?origin=scopusAI

77. Leveraging Explainable AI and Sarcasm Features for Improved Cyberbullying Detection in Multilingual Settings Mahmud, T., Ptaszynski, M., Masui, F. 2024 IEEE Digital Platforms and Societal Harms, DPSH 2024, 2024 https://www.scopus.com/pages/publications/85213956360?origin=scopusAI

78. Dialect Bias in Arabic Toxicity Detection Under Controlled Distributional Shifts Alamri, M. Electronics (Switzerland), 2026 https://www.scopus.com/pages/publications/105048326793?origin=scopusAI

79. Fairness in Language Models Beyond English: Gaps and Challenges Ramesh, K., Sitaram, S., Choudhury, M. EACL 2023 - 17th Conference of the European Chapter of the Association for Computational Linguistics, Findings of EACL 2023, 2023 https://www.scopus.com/pages/publications/85159851850?origin=scopusAI

80. Automatic classification of participant roles in cyberbullying: Can we detect victims, bullies, and bystanders in social media text? Jacobs, G., Van Hee, C., Hoste, V. Natural Language Engineering, 2022 https://www.scopus.com/pages/publications/85096446182?origin=scopusAI